



Team Name:

MEMORA

Team Member's University:

UNIVERSITI SAINS MALAYSIA

Team Member's Name

No.	NAME	Matric Number
1.	LEE TECK QING	23301010
2.	LIM TIAN YANG	23301210
3.	LIM THER HOW	23302867
4.	LIEW JIAN SHENG	23301095
5.	YAP TIAN SHEN	23301794

Title:

MEMORA

"Digital Infrastructure for Ageing in Place:

Integrating Safety, Independence, and Reminiscence Therapy."

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1 Executive Summary

Malaysia faces an imminent "Care Crisis." By 2030, the elderly population will exceed 15%, yet the nation faces a severe shortage of affordable care facilities. The current reliance on institutionalization ("nursing homes") is economically unsustainable for the government and culturally difficult for families.

MEMORA proposes a scalable solution to bridge this gap. We are developing a **Community Safety Infrastructure** designed to enable "**Ageing in Place**"—allowing seniors to live safely in their own homes for longer, thereby reducing the burden on public health facilities.

Unlike standard trackers which prioritize surveillance, MEMORA is built on the ***Kaigo* (Supported Independence)** philosophy. It combines **Physical Safety Nets** (Fall Detection, Geo-fencing) with **Psychosocial Support** (Digital Reminiscence Therapy) to protect both the physical body and the dignity of the user.

We aim to validate this through a **Public-Private Partnership (PPP/PFI) Model**, demonstrating how preventative digital infrastructure can be funded sustainably by sharing costs between public subsidies, private insurance, and family contributions.

2 Introduction: The Demographic Context

The Demographic Reality

The "Silent Epidemic" of elderly isolation is no longer just a social issue; it is an economic urgency. According to the Department of Statistics Malaysia (DOSM), Malaysia is rapidly becoming an aged nation. However, infrastructure growth has not kept pace. Estimates indicate that between 7.4% and 10% of elderly Malaysians live alone, often without the "middle ground" support needed between total independence and full-time nursing care.

The Core Problem: Safety vs. Dignity

Currently, families face a binary choice: leave parents at home at risk of unnoticed accidents (such as the 80 solitary deaths recorded in Perlis alone), or move them to facilities that may compromise their autonomy. Existing market solutions—smartwatches and trackers—often fail because they are designed as "surveillance tools," leading to low adoption rates among seniors who feel their dignity is being compromised.

The MEMORA Solution

MEMORA enters the market not as a gadget, but as a **support system**. Our mission is to operationalize "**Community Care**" by connecting the elderly to a safety network of family members and professional caregivers.

Our approach rests on two pillars:

1. **Supported Independence:** Using unobtrusive sensors (Heart Rate, SpO2, Accelerometers) to detect emergencies like falls or wandering, allowing seniors to move freely within their community without fear.

2. **Digital Legacy (Reminiscence Therapy):** Addressing the mental health crisis of isolation through our "Legacy Diary" feature. By integrating voice messages and memories from family into the device, we turn a safety tool into an emotional connection, improving user compliance and mental well-being.

Strategic Vision

Our vision is to establish MEMORA as the standard **Digital Infrastructure for Ageing in Place** in Malaysia. By leveraging a sustainable PFI business model, we aim to ensure that no elderly person is left unseen, unheard, or forgotten, regardless of their economic background.

3 Problem Statement: The Infrastructure Gap

3.1 The Macro Challenge: Demographics vs. Capacity

Malaysia is undergoing a rapid demographic transition. According to the Department of Statistics Malaysia (DOSM), the elderly population (aged 60 and above) is projected to exceed **15% of the total population by 2030**, officially granting the nation "Aged Status". However, the physical infrastructure required to support this shift is critically insufficient.

With only **393 registered elderly care centres** and 26 nursing homes currently available, the traditional model of institutionalizing the elderly is mathematically and economically impossible to scale at the required speed. The "waiting list" reality for facilities forces a reliance on **Ageing in Place** (staying at home), yet the majority of Malaysian homes lack the safety infrastructure to support this.

In this proposal, we refer to the elderly user not as a 'patient,' but as a '**Memorian**'. This terminology reflects our core philosophy: they are not just subjects to be monitored, but individuals whose legacy and independence are being preserved.

3.2 The "Silent Epidemic" of Isolation

The social consequence of this infrastructure gap is the rise of the "Silent Epidemic"—elderly individuals living, and dying, in total isolation.

- **Solitary Living:** DOSM estimates indicate that between **7.4% and 10%** of elderly Malaysians live alone.
- **Unnoticed Mortality:** The risks are lethal. In Perlis alone—a state with a small population—there were **80 recorded cases** of elderly individuals dying alone between 2022 and August 2024.
- **The Care Void:** These individuals face unnoticed accidents, deteriorating health conditions, and a total lack of real-time assistance.

3.3 The "Missing Middle" in Technology

While families wish to support their aging parents, they face a "Usage Barrier." Current market solutions fail to address the specific needs of the Malaysian elderly, particularly those in the B40 group or suffering from cognitive decline:

1. **Complexity:** Smartwatches are often too complex for seniors with low digital literacy, leading to abandonment of the device.
2. **Stigma:** Clinical trackers feel like "patient tags," stripping the user of dignity.
3. **Cost:** High-end monitoring devices are financially inaccessible to the mass market.

Conclusion: There is a critical lack of **decentralized safety infrastructure** that bridges the gap between total independence and institutional care. MEMORA aims to fill this void by providing the "Digital Safety Net" required to make Ageing in Place a safe, viable, and dignified reality.

4 Proposed Solution: MEMORA Infrastructure

4.1 Concept Overview: A Digital Safety Net

MEMORA is a decentralized wearable infrastructure designed to enable safe **Ageing in Place** for the Malaysian elderly. Unlike standard consumer smartwatches, MEMORA is purpose-built to bridge the gap between independent living and professional care. It operates on a **"Supported Independence"** model: providing the safety data of a medical facility while the user remains in the comfort of their own home.

4.2 Pillar 1: The Safety Ecosystem (Physical Security)

The hardware is designed as an unobtrusive "Safety Net" that runs passively in the background, minimizing the need for user interaction (lowering the barrier for dementia patients).

- **Vital Sign Trending:** The device continuously monitors baseline metrics, including heart rate, body temperature, and motion activity. Rather than replacing doctors, it provides trend analysis to alert caregivers of sudden anomalies.
- **Fall Detection & Response:** Utilizing a calibrated accelerometer and gyroscope, the device detects sudden impacts (falls). This triggers an immediate alert protocol: first to the primary caregiver, and if unanswered, escalating to emergency services or a partnered hospital.
- **Geofencing for Autonomy:** To support seniors with early-stage cognitive decline who wish to walk outdoors, MEMORA utilizes GPS tracking. Caregivers can define "Safe Zones" (e.g., the neighbourhood block). This allows the user freedom of movement while ensuring immediate alerts if they wander too far.
- **Sleep & Recovery Analysis:** The device tracks sleep stages (Light, Deep, REM) to assist caregivers in assessing the senior's recovery quality and detecting sleep disturbances common in ageing.

4.3 Pillar 2: Psychosocial Support (**Kaigo & Legacy**)

To combat the "patient stigma" often associated with safety wearables, MEMORA integrates **Digital Reminiscence Therapy** tools directly into the infrastructure. Standard trackers feel like surveillance. MEMORA adopts a ***Kaigo-centric approach***, prioritizing the user's dignity and emotional health through Reminiscence Therapy (The Legacy Diary)."

- **The Legacy Diary:** This feature transforms the device from a "tracker" into a "connection tool." By recording their stories, the **Memorian** leaves a digital inheritance for their family. This shifts the dynamic from 'surveillance' to 'storytelling,' empowering the Memorian to feel like an active contributor to the family's history rather than a burden. Families can upload voice notes and memories to the device. This function serves a dual purpose:
 1. **Mental Health:** It combats social isolation by keeping the senior connected to family voices.
 2. **Compliance:** Seniors are more likely to wear the device to hear messages from grandchildren, increasing safety compliance without coercion.
- **Medication Autonomy:** Vibration and light-based reminders help users manage their own medication schedules, fostering self-reliance rather than dependence.

4.4 Technical & Security Architecture

- **Caregiver Sync Dashboard:** All health and location data is synced to a mobile application that supports multiple user roles, allowing a "Care Circle" (children, grandchildren, and hired nurses) to monitor the same user.
- **Data Privacy & Encryption:** Recognizing the sensitive nature of health data, the MEMORA ecosystem runs on a private server with end-to-end encryption (AES-256), ensuring full PDPA compliance and user privacy.
- **Sustainable Design:** To align with long-term sustainability goals, the device utilizes rechargeable Li-Po batteries, reducing electronic waste compared to disposable battery trackers.

5 Strategic Analysis

5.1 SWOT Analysis

A strategic evaluation of the MEMORA infrastructure within the current Malaysian aged-care landscape.

STRENGTHS (Internal)	WEAKNESSES (Internal)
1. Holistic Monitoring Infrastructure: Unlike basic trackers, MEMORA integrates vital sign trending (heart rate/SpO2) with physical safety (fall detection) and location tracking.	1. Digital Literacy Barriers: Reliance on a smartphone app for the caregiver interface may hinder adoption among older, less tech-savvy guardians.
2. Digital Legacy Integration: The inclusion of the "Legacy Diary" provides a unique emotional value proposition, differentiating the device from purely clinical competitors.	2. Power Dependencies: The inclusion of GPS and continuous monitoring places strain on battery life, necessitating disciplined charging routines which may be burdensome for dementia patients.
3. Real-Time Response: Immediate alerts allow for rapid intervention by family or professional caregivers, reducing the "found dead alone" risk.	3. Cost vs. Accessibility: High hardware costs may alienate the B40 demographic without subsidy or PFI intervention.

OPPORTUNITIES (External)	THREATS (External)
1. Institutional Integration: Partnerships with nursing homes to automate resident monitoring can reduce staff workload and improve response times.	1. Market Saturation: The influx of generic health wearables (e.g., Xiaomi, Apple) threatens market share if differentiation is not clear.
2. The Dementia Care Gap: High demand for tracking solutions for patients with cognitive impairments who are prone to wandering.	2. Cultural Resistance: Elderly populations may view wearable tech as invasive ("surveillance") rather than supportive, leading to non-compliance.
3. PFI Potential: Opportunity to position MEMORA as a government-subsidized preventative tool to lower public hospital admissions.	3. Economic Volatility: Recessionary pressures typically reduce discretionary spending on health devices, impacting direct-to-consumer sales.

5.2 Competitive Landscape

The current market is bifurcated into "Clinical Medical Devices" and "Consumer Gadgets," neither of which fully serves the Ageing-in-Place need.

1. **Clinical Grade (e.g., Empatica EmbracePlus, CardiacSense):** These devices offer FDA-grade monitoring but suffer from "Patient Stigma." They look medical, feel clinical, and lack emotional engagement features, leading to low daily compliance among seniors who do not wish to feel "sick."
2. **Consumer Wearables (e.g., Movano Evie, Smartwatches):** While aesthetically pleasing, many suffer from short battery life (often <3 days) or require complex OS integration (iOS 16+) that alienates the elderly demographic. They also lack specific "Safety Net" features like dedicated SOS flows for caregivers.

The Gap: There is currently no dominant player providing a "**Dignity-First**" **Safety Wearable** that combines safety tracking with the emotional incentives required for long-term adoption.

5.3 Value Proposition (The MEMORA Advantage)

MEMORA bridges the gap between clinical utility and human connection.

1. **Hybrid Data Ecosystem:** We capture not just biological data (heart rate, sleep, motion) but also "Contextual Safety Data" (location, fall history) to paint a complete picture of the user's well-being.
2. **Actionable Analytics:** Unlike passive trackers, our dashboard provides caregivers with actionable insights—alerting them only when trends deviate from the norm, preventing "alert fatigue".
3. **The "Compliance Engine" (Heritage of Memory):** Our strongest differentiator is the **Legacy Diary**. By allowing users to record memories and receive messages from family, we transform the device into a sentimental keepsake. This "Emotional Infrastructure" ensures the elderly *want* to wear the device, solving the critical industry problem of abandonment/non-compliance.

6 Sustainability & PFI Adoption Strategy

6.1 Product Positioning: "More Than a Tracker"

To ensure high adoption rates, MEMORA is positioned not as a medical surveillance device, but as an **All-in-One Health Guardian & Family Keepsake**. This dual positioning addresses the two primary barriers to adoption:

1. **Functional:** Providing advanced safety sensors (Fall Detection, SpO2, GPS).
2. **Emotional:** Using the **Digital Legacy Diary** to create a sentimental bond between the user and the device, ensuring the elderly *want* to wear it rather than feeling forced.

We plan to consult with **Kaigo-certified professionals** to ensure our fall detection protocols respect the user's autonomy and do not cause unnecessary embarrassment.

6.2 Funding Model: The PFI & Cross-Subsidy Framework

To ensure financial sustainability while remaining accessible to all income brackets, we propose a dual-funding strategy leveraging a **Public-Private Partnership (PPP)** model, reinforced by an internal cross-subsidy protocol.

A. The Private Market (B2C Model - M40/T20) For middle-to-high income families, we operate on a standard hardware-plus-subscription model to generate primary revenue:

- **Hardware:** Tiered pricing starting at **RM 699** (Gen 1) to **RM 1199** (Gen 1 Pro).
- **Service:** Monthly subscription tiers (Silver/Gold/Diamond) ranging from **RM 40** to **RM 100**.

B. The "Social Impact" Cross-Subsidy (The MEMORA Commitment) To demonstrate our commitment to the national care agenda, MEMORA implements a mandatory **Profit Reinvestment Protocol**:

- **20% Cross-Subsidization:** We pledge that **20% of all net profits** generated from the M40/T20 consumer sales will be strictly ring-fenced to subsidize the hardware costs for B40 families.

- **Impact:** This internal funding mechanism reduces the reliance on government grants, making the project fiscally attractive to state partners who only need to "top up" the remaining cost rather than funding the entire device.

C. The Social Infrastructure Model (B2G - B40) For the remaining cost gap for low-income seniors, we propose a PFI structure:

- **Government Role:** Matching grants to cover the remaining Capital Expenditure (CapEx) after the Memora subsidy is applied.
- **Private Insurance:** Partnering with PERKESO or private insurers to cover the monthly OpEx (monitoring fee) as a preventative measure against hospitalization.
-

6.3 Deployment Strategy (B2B & B2G)

We move beyond standard retail ("Distribution") to a strategic deployment model focused on integration with care providers.

1. **Institutional Partnership (B2B):** Direct integration with nursing homes and eldercare centres. MEMORA provides a centralized dashboard for staff to monitor multiple residents simultaneously, increasing operational efficiency.
2. **Clinical Discharge Program:** Partnering with hospitals to offer MEMORA to elderly patients upon discharge, ensuring they have a "safety net" during the critical recovery period at home.
3. **Community Outreach (NGOs):** Collaborating with NGOs and CSR initiatives to deploy subsidized units into rural areas where physical access to clinics is limited.

6.4 Engagement & Adoption Strategy

The challenge in aged care is not just selling the device, but ensuring **compliance** (daily usage).

- **The "Trojan Horse" Approach:** We market the device to the *elderly* as a connection to their grandchildren (via the Legacy Diary), while marketing the *safety* features to the *adult children*. This ensures the senior wears it for love, while the family buys it for peace of mind.
- **Community Education:** Conducting workshops with ***Kaigo*** certified trainers to educate families on how digital tools can support, rather than replace, human care.
- **Referral Ecosystem:** Implementing a referral program where families can earn lifetime cloud backup or discounts, encouraging community-led growth.

7 Implementation & Operational Roadmap

7.1 Strategic Phasing Overview

Our roadmap is designed to validate the "Ageing in Place" efficacy before attempting mass scale. We prioritize **Clinical Validation** over rapid commercialization to ensure the safety infrastructure is robust enough for the PFI model.

Phase 1: Infrastructure & R&D (Months 1-4)

- **Objective:** Finalize hardware tooling and secure server architecture.
- **Key Activity:** Establishing the **OEM Partnership** with Tier-1 manufacturers capable of medical-grade sensor integration and Li-Po safety standards.
- **Kaigo Integration:** Consulting with professional care providers to calibrate the "Fall Detection" sensitivity to minimize false positives, ensuring dignity for the user.

Phase 2: The "Evidence-Based" Pilot Program (Months 5-7)

- **Objective:** Proof of Concept (PoC) with **n=50** users.
- **Partnership:** We will deploy the first cohort in partnership with a local NGO or nursing facility rather than direct-to-consumer sales.
- **Data Collection:** This phase focuses on gathering data on **"Response Time"** (how fast caregivers react to alerts) and **"Compliance"** (do seniors actually wear it because of the Legacy Diary?). This data is required to pitch to insurers for the PFI model.

Phase 3: Commercial Rollout & PFI Integration (Months 8-12)

- **Objective:** Scale to 1,000+ units and activate the B40 Cross-Subsidy.
- **Activity:** Launching the B2C store to generate the profit margin required to fund the B40 subsidy. Simultaneously presenting Pilot Data to government stakeholders for PFI grants.

7.2 Supply Chain & Quality Assurance

To maintain the integrity of a safety device, we eschew the traditional retail model for a direct-to-infrastructure approach.

- **Manufacturing:** Contracted with established OEMs for PCB and mechanical enclosure fabrication.
- **Logistics:** Partnering with a 3PL provider in Kuala Lumpur for centralized fulfilment, ensuring replacement units can reach users within 24 hours—critical for a safety product.
- **Sustainability:** Implementation of a "Hardware Recycling Program" to refurbish devices from the M40 market for redeployment in B40 communities, extending the product lifecycle.

7.3 Technology & Security Roadmap

Given the sensitivity of health and location data, our software architecture prioritizes privacy compliance (PDPA).

- **Mobile Interface:** Developed on **React Native** for cross-platform accessibility (iOS/Android).
- **Backend Infrastructure:** Utilizing **Node.js + AWS** (Lambda/DynamoDB) for scalable, serverless data management.
- **Security Protocol:** All "Legacy Diary" voice data and location history are secured with **AES-256 end-to-end encryption**. Only the authorized "Care Circle" (family and linked professional caregivers) possess the decryption keys.

7.4 Final Evaluation & KPI Framework

At the conclusion of the first operational year, project success will be evaluated against a "Triple Bottom Line" framework (Financial, Social, and Operational Impact), prioritizing evidence-based outcomes over pure commercial volume.

1. Operational Safety Metrics (The *Kaigo* Standard)

- **Mean Time to Resolution (MTTR):** Measuring the average time elapsed between a "Fall Alert" and "Caregiver Response." Our target is <5 minutes.
- **False Positive Rate:** Tracking the accuracy of the fall detection algorithm to ensure user dignity is maintained (minimizing false alarms that cause embarrassment).

2. Social Impact Metrics (The PFI Standard)

- **Equitable Access:** Successfully directing the **20% Profit Subsidy** to fully fund hardware for at least 200 B40 families in the pilot phase.
- **Adoption & Compliance:** Achieving a **Retention Rate of >60%** among users, validating that the "Legacy Diary" feature effectively combats device abandonment.

3. Financial Health

- **PFI Viability:** Gathering sufficient data on "Hospitalization Prevention" to present a case study to private insurers for long-term partnership.

8 Financial Feasibility

8.1 Start-Up Capital Strategy

To launch the MEMORA infrastructure, we require an initial seed capitalization of **RM 347,200.00**. Unlike traditional startups that burn capital on aggressive marketing, our allocation prioritizes **R&D and Evidence Generation** to validate the PFI model for government adoption.

Capital Allocation Breakdown:

- **Hardware & Tooling (45%):** Manufacturing the initial pilot batch (1,200 units) including tooling for the "Shock Absorber" casing (Polycarbonate + TPU) crucial for durability.
- **App & Infrastructure (22%):** Developing the secure, encrypted cloud architecture and the "Legacy Diary" backend.
- **Pilot & Validation (10%):** Funding the "n=50" pilot program to gather the "Response Time" data required for government grant applications.
- **Operations & Legal (23%):** IP registration, server hosting, and initial staffing.

Funding Mix: We propose a hybrid funding structure to de-risk the venture:

1. **Founder Contribution:** RM 52,500 (15%) – Demonstrating "Skin in the Game".
2. **Innovation Grants:** RM 52,500 (15%) – Targeting "Ageing Nation" grants (e.g., MOSTI/Cradle).
3. **Angel/Strategic Investment:** RM 245,000 (70%) – Seeking partners aligned with the "Care Economy" vision.

8.2 Unit Economics & Technical Optimization

We have optimized our Bill of Materials (BOM) to prioritize **Safety Reliability** (Battery Life) over component cost.

The Hardware Upgrade: We have replaced the originally proposed ESP32 microcontroller with the **Nordic nRF52840**.

- **Reason:** The ESP32 is power-intensive (Wi-Fi based), risking battery failure in <24 hours. The Nordic chip utilizes **Bluetooth Low Energy (BLE 5.0)**, extending battery life to **5–7 days**. This reduces the "charging burden" on elderly users, a critical factor for compliance.

Manufacturing Cost Structure (Revised):

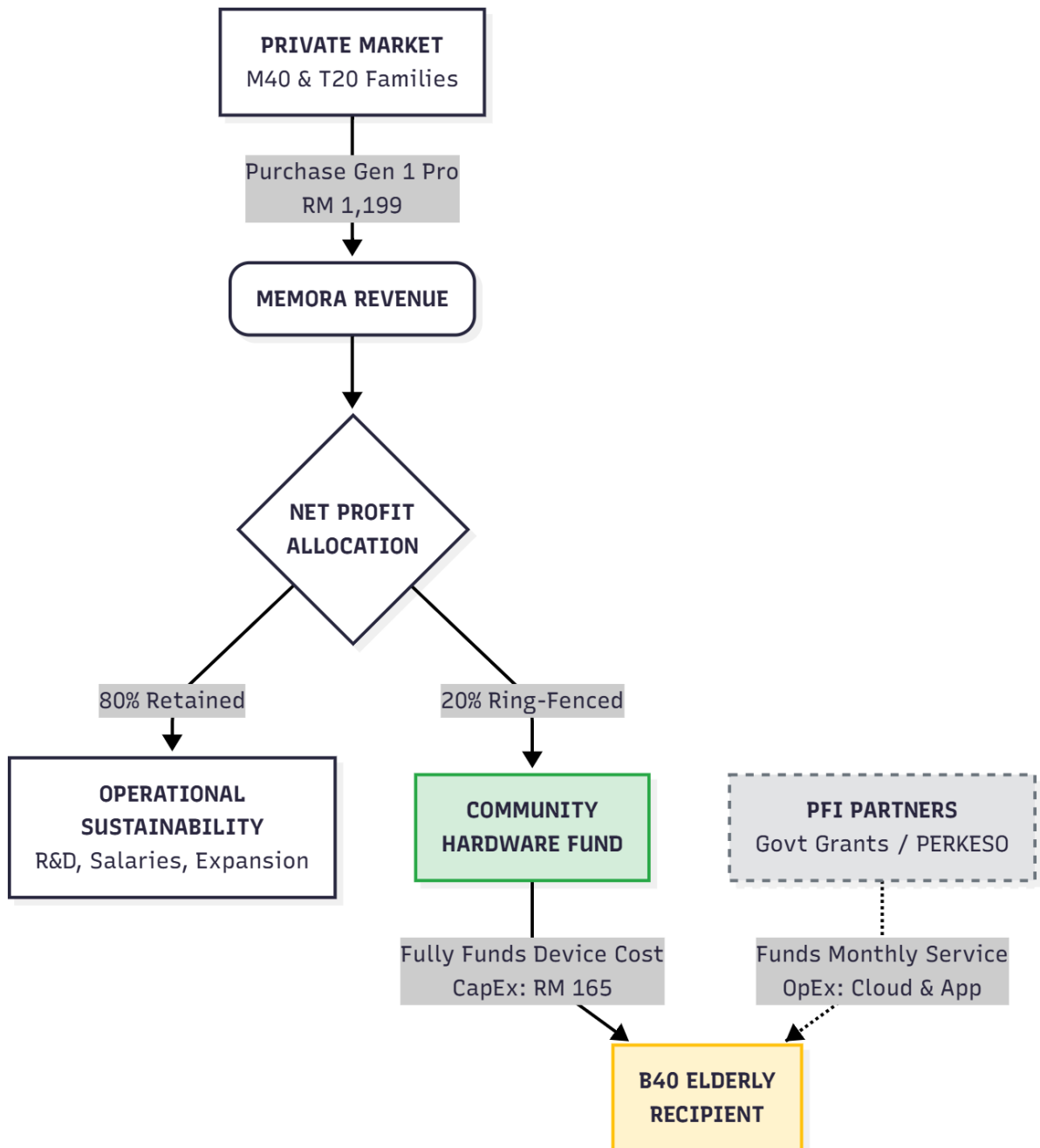
- **Core Unit (Gen 1): RM 165.00** (Adjusted for Nordic SoC).
 - *Key Components:* Nordic nRF52840 SoC, MAX30102 (Vital Sensor), MPU6050 (Fall Detection).
- **Pro Unit (Gen 1 Pro): RM 195.00.**
 - *Includes:* Expanded onboard memory for offline Legacy Diary storage.

Pricing Strategy (The Cross-Subsidy Engine):

- **Private Market Price:** RM 699 (Gen 1) / RM 1,199 (Pro).
- **Subscription:** RM 40 - RM 100/month.
- *Note:* The healthy gross margin on the Pro unit (>80%) is not purely profit; it acts as the funding engine for our B40 Social Program.

8.3 The "PFI" Sustainability Model (Revenue & Impact)

We move away from "Sales Volume" as the only metric and focus on **Ecosystem Sustainability**. Our model utilizes profit from the M40/T20 market to subsidize the B40 demographic, reducing the reliance on perpetual government funding.



The "20% Cross-Subsidy" Protocol:

1. **Revenue Generation:** Sales from the affluent market (Gen 1 Pro) generate a Gross Margin of ~RM 1,000 per unit.
2. **The Ring-Fence: 20% of net profit** from these sales is strictly directed to a "Community Hardware Fund."
3. **Business Viability:** Even after dedicating 20% to social impact, MEMORA retains **80% of profits** for R&D and Operations. This allocation is strategic: it qualifies MEMORA as a "Social Enterprise," unlocking government matching grants that far exceed the cost of the subsidy.
4. **PFI Deployment:** The fund covers the hardware cost for B40 seniors, meaning the government or insurance partner only needs to cover the monthly cloud cost (OpEx).

8.4 Projected Financial Trajectory

- **Year 1 (Validation):**
 - **Target:** Deployment of 1,200 pilot units (Gen 1 & Pro).
 - **Focus:** Securing a "Subscription Retention Rate" of >50%. We value recurring revenue (Infrastructure) over one-time hardware sales.
 - **Projected Revenue:** ~RM 4.6 Million (assuming full sell-through of pilot inventory and subscriptions).
- **Year 2-3 (Scale & PFI Integration):**
 - **Target:** 6,500 units.
 - **Strategy:** Institutional bulk-deployment (Nursing Homes & NGOs).
 - **Impact:** By Year 3, the ecosystem aims to support over 150 care facilities, validating MEMORA as a critical component of Malaysia's national aged-care infrastructure.

9 Technical Specifications (Prototype)

9.1 Industrial Design: The "Dignity-First" Form Factor

To address the high rate of device abandonment among the elderly, MEMORA avoids the "medical tag" aesthetic. The prototype features a modular core housed in three interchangeable casings (**Retro, Luxury, Sport**).

- **Materiality:** The outer shell utilizes **Polycarbonate (PC) with a TPU coating**. This combination is hypoallergenic for long-term wear and provides high impact resistance, crucial for surviving patient falls.
- **Shock Absorption:** An internal layer of **EVA Foam** acts as a damper to protect the PCB from sudden impacts, ensuring the "Fall Detection" sensors remain calibrated even after rough usage.

9.2 Internal Architecture & Layering

The device is constructed using a "Sandwich Architecture" to minimize thickness while maximizing protection.

- **L1 (Outer):** Polycarbonate Shell (Hard protection).
- **L2 (Mid):** Microporous Silicone Sheet (Heat insulation to prevent skin irritation from battery warmth) .
- **L3 (Core):** Component Layer (PCB, Sensors, Battery).

9.3 System Architecture & Components

We have optimized the Bill of Materials (BOM) to prioritize battery longevity (safety reliability) over standard connectivity.

Component	Part Number	Function & Rationale
Microcontroller (SoC)	Nordic nRF52840	Upgrade from ESP32: Utilizes Bluetooth Low Energy (BLE 5.0) to extend battery life to 5–7 days, ensuring the safety net remains active longer.
Vital Sensor	MAX30102	Optical sensor for Heart Rate & SpO2 trending. Provides "Baseline Deviation" alerts (e.g., sudden drop in oxygen).
Fall Sensor	MPU6050	6-Axis Accelerometer & Gyroscope. Calibrated to distinguish between daily motion and hard impact (falls).
Location	NEO-6M GPS	Passive location tracking. Activates primarily when "Geofence Breach" is detected to conserve power.
Thermal	MLX90614	IR Thermometer to track body temperature trends (fever detection).

9.4 Safety Mechanisms: The "Secure Clasp"

To support users with early-stage dementia who may attempt to remove the device (wandering risk), we have designed a **Tamper-Resistant Clasp**.

- **Mechanism:** A modified "Cable Tie" concept that requires a specific dual-finger motion to release.
- **Kaigo Benefit:** This ensures the device stays on during episodes of confusion without feeling like a "shackle" or causing physical discomfort.

9.5 Emergency Response Protocol (Logic Flow)

The system is programmed with a "Tiered Escalation" protocol to minimize false alarms while ensuring safety.

1. **Event Trigger:** Fall detected OR Geofence Breach.
2. **Local Verification:** Device vibrates/buzzes. If user presses button, alert is cancelled (False Positive prevention).
3. **Tier 1 Alert:** If no cancel, alert sent to **Primary Caregiver (App)** via Cloud.
4. **Tier 2 Escalation:** If Caregiver does not acknowledge within 3 minutes, the system triggers the **Professional Support Link** (if subscribed) or Emergency Services.

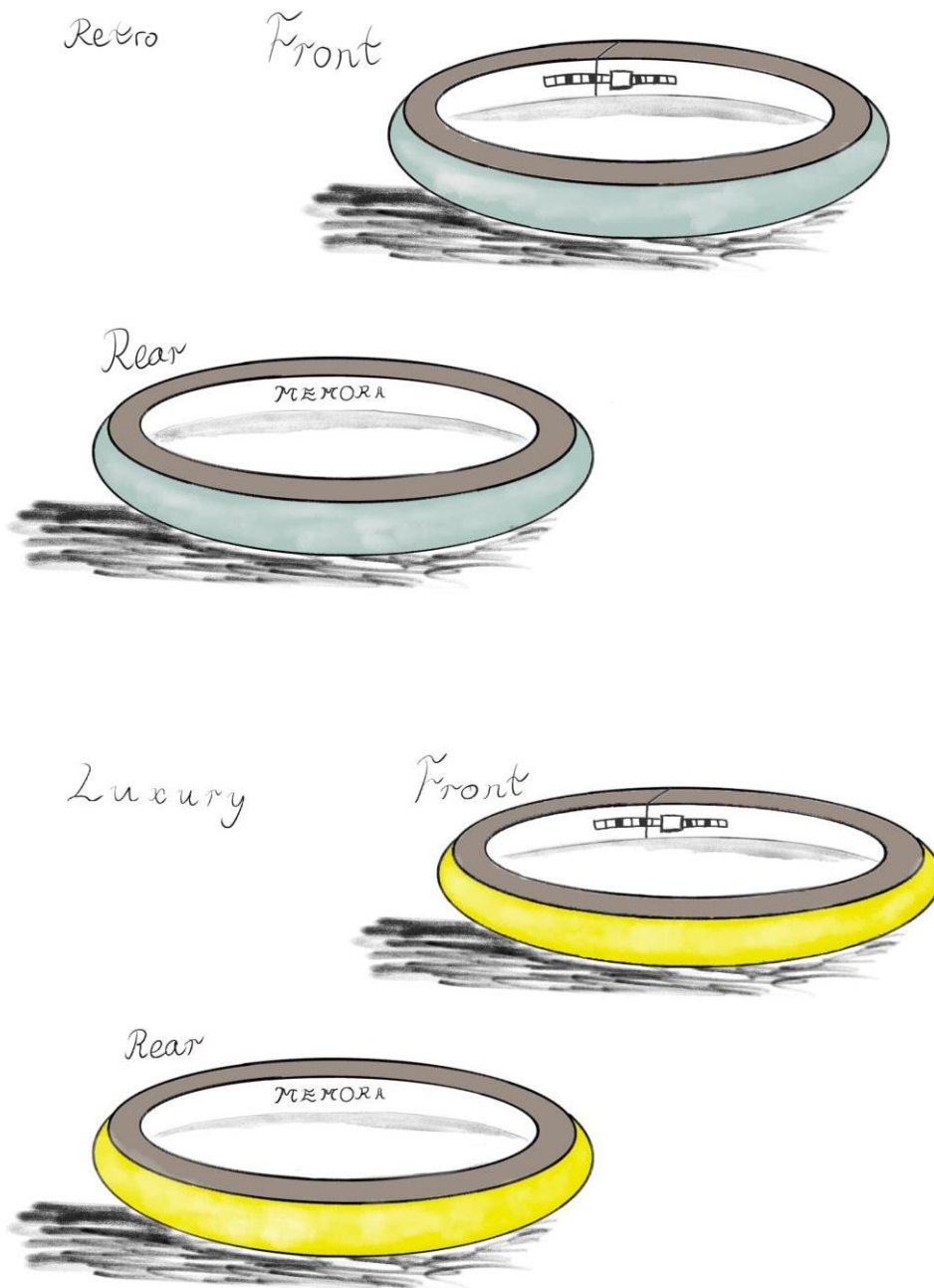
9.6 User Interface (App Ecosystem)

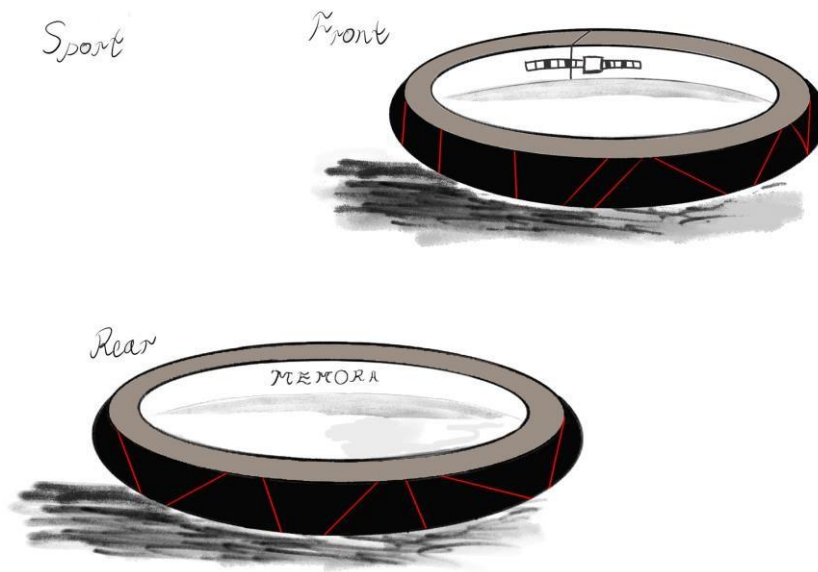
The App is designed for the *Caregiver*, not the senior, reducing the digital literacy barrier.

- **Dashboard:** Displays "Green/Red" status for Vitals and Location.
- **Legacy Diary Hub:** A dedicated interface for families to upload voice notes and photos. These are synced to the bracelet to play at scheduled times (e.g., medication time), using the voice of a loved one to encourage compliance.

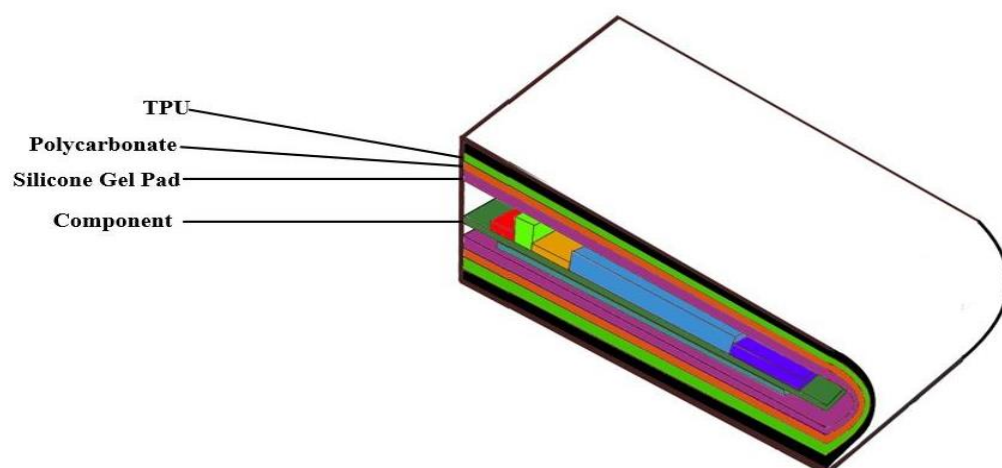
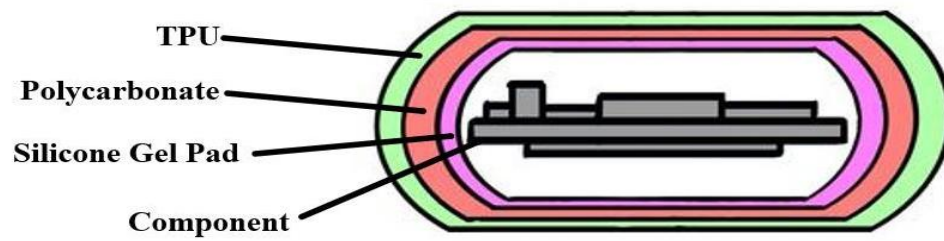
9.7 Memora Design

- **Model**



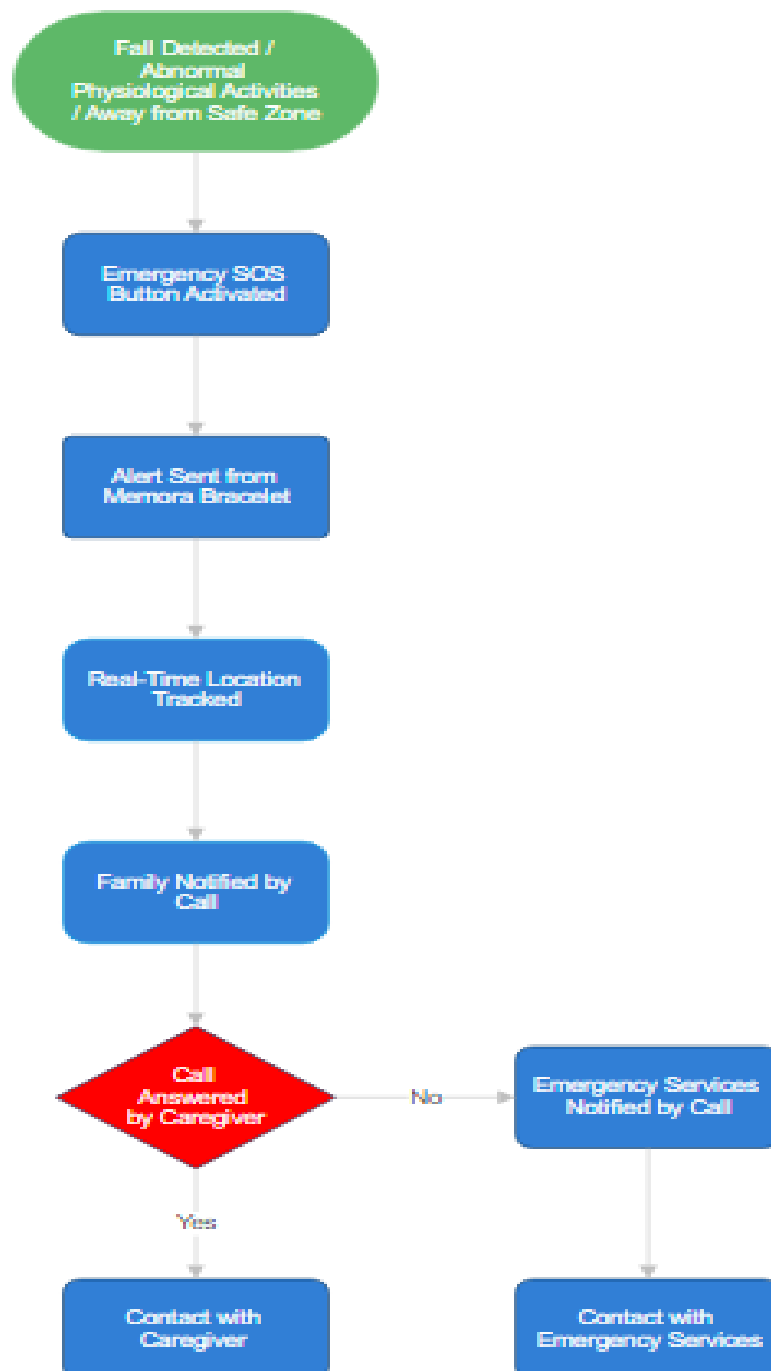


- **Cross Sectional Drawing**

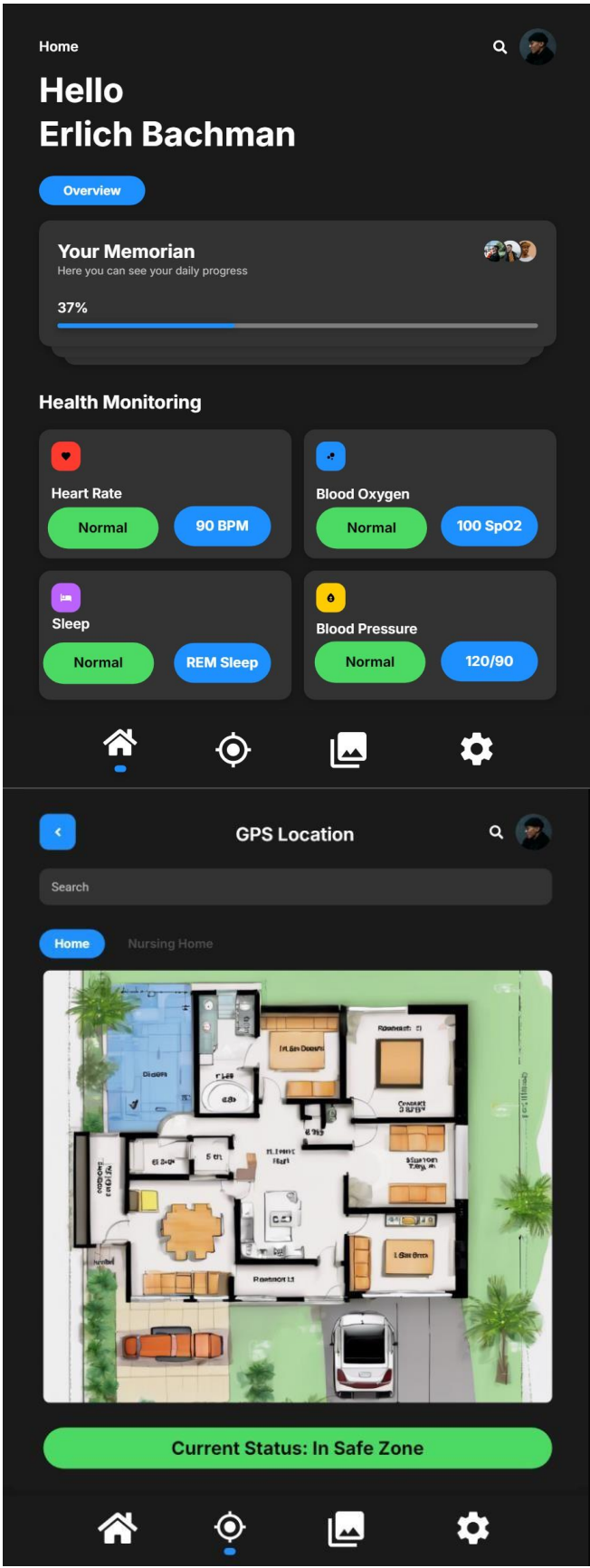


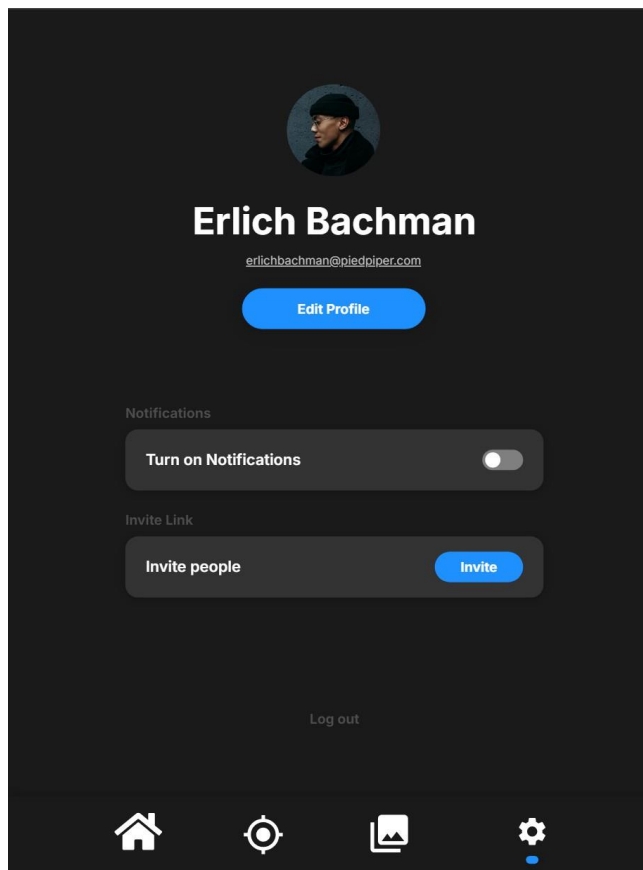
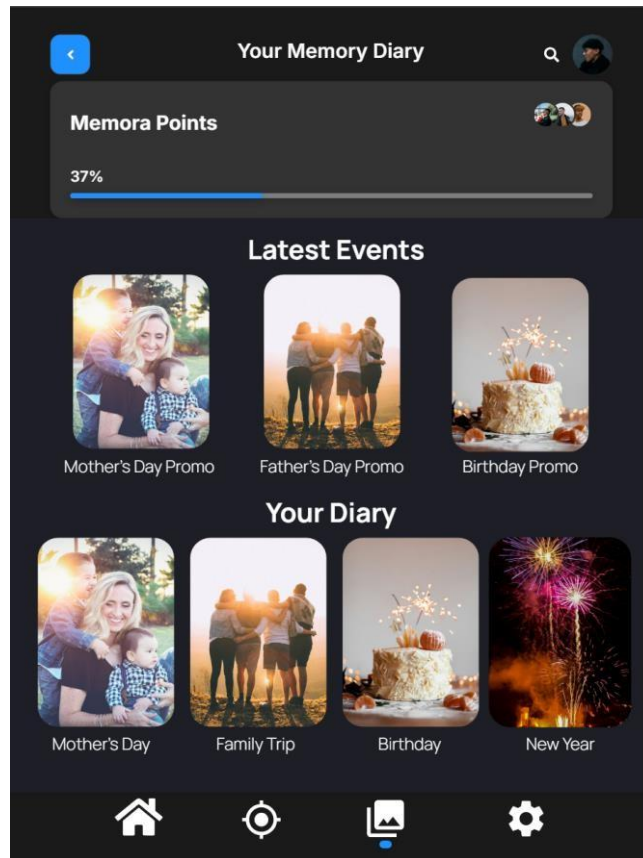
- Emergency Flow System

Emergency SOS Flow Diagram



- APP Design





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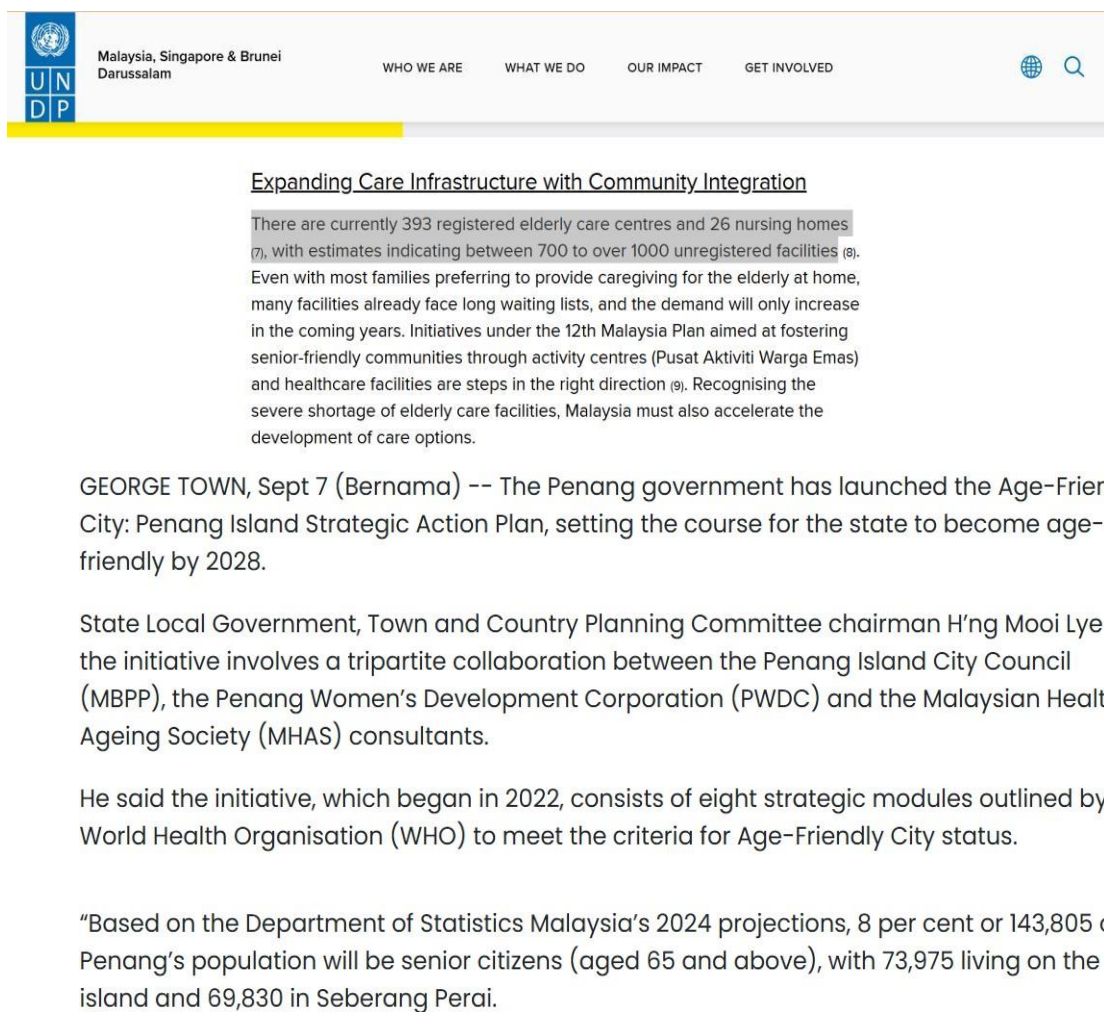
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11 Appendix

11.1 Appendix A: Validation of the Infrastructure Gap – Only 393 registered centres for a growing aged population.



The screenshot shows the header of a UNDP report for Malaysia, Singapore & Brunei Darussalam. The navigation bar includes links for 'WHO WE ARE', 'WHAT WE DO', 'OUR IMPACT', and 'GET INVOLVED'. The main content area is titled 'Expanding Care Infrastructure with Community Integration' and contains the following text:

There are currently 393 registered elderly care centres and 26 nursing homes (7), with estimates indicating between 700 to over 1000 unregistered facilities (8). Even with most families preferring to provide caregiving for the elderly at home, many facilities already face long waiting lists, and the demand will only increase in the coming years. Initiatives under the 12th Malaysia Plan aimed at fostering senior-friendly communities through activity centres (Pusat Aktiviti Warga Emas) and healthcare facilities are steps in the right direction (9). Recognising the severe shortage of elderly care facilities, Malaysia must also accelerate the development of care options.

GEORGE TOWN, Sept 7 (Bernama) -- The Penang government has launched the Age-Friendly City: Penang Island Strategic Action Plan, setting the course for the state to become age-friendly by 2028.

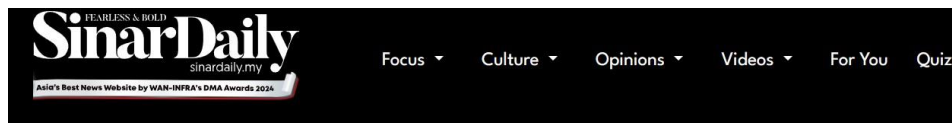
State Local Government, Town and Country Planning Committee chairman H'ng Mooi Lye said the initiative involves a tripartite collaboration between the Penang Island City Council (MBPP), the Penang Women's Development Corporation (PWDC) and the Malaysian Healthy Ageing Society (MHAS) consultants.

He said the initiative, which began in 2022, consists of eight strategic modules outlined by the World Health Organisation (WHO) to meet the criteria for Age-Friendly City status.

"Based on the Department of Statistics Malaysia's 2024 projections, 8 per cent or 143,805 of Penang's population will be senior citizens (aged 65 and above), with 73,975 living on the island and 69,830 in Seberang Perai.

Context: A compilation of data highlighting the critical supply-demand disparity in Malaysian aged care. The UNDP report confirms there are only **393 registered care centres** nationwide, a capacity that is statistically insufficient to support the projected demographic shift (e.g., **143,805 seniors in Penang alone** as cited by DOSM). This evidence validates MEMORA's core premise: that "Ageing in Place" is not just a preference, but an economic necessity due to the severe shortage of institutional beds.

11.2 Appendix B: Evidence of the 'Silent Epidemic' – 80 solitary deaths in Perlis (2024).



A recent Sinar report revealed a growing number of such cases.

In Perlis alone, a state with fewer than 300,000 residents, 80 solitary deaths were recorded between 2022 and August 2024.



According to Persatuan Sains Sosial Malaysia (PSSM) special advisor Professor Emeritus Datuk Dr Abdul Rahman Embong, United Nations (UN) data from 2021 ranked Malaysia as the third-highest Asean country in terms of unmarried individuals, at 44.7 per cent.

He warned that Malaysia's rapidly ageing population is on track to becoming an aged nation before achieving developed status, leading to an increasing number of people living alone.

Context: Data highlighting the urgent mortality risk associated with elderly isolation. The report notes **80 recorded solitary deaths** in Perlis (a state with <300k population) between 2022 and 2024. This statistic underscores the critical "Infrastructure Void" where elderly individuals pass away unnoticed due to a lack of real-time safety monitoring, validating the necessity for MEMORA's 24/7 activity tracking.

11.3 Appendix C: Media Evidence of the "Silent Epidemic" Source: *The Star* (10 July 2023).

 Elderly man dies at home, alone

NATION
Monday, 10 Jul 2023

Related News



NATION 07 Apr 2025
'Better to keep watch
than regret later if
something goes missing'

MUAR: He was found lying motionless on a bed in the living room of his house where he lived alone.

It was the third day of Hari Raya Aidilfitri, an occasion which he should have been celebrating.

Instead, his neighbours detected an odour coming from his home. The 75-year-old man was believed to have been dead for at least three days before his decomposed body was found at about 11pm on April 24.

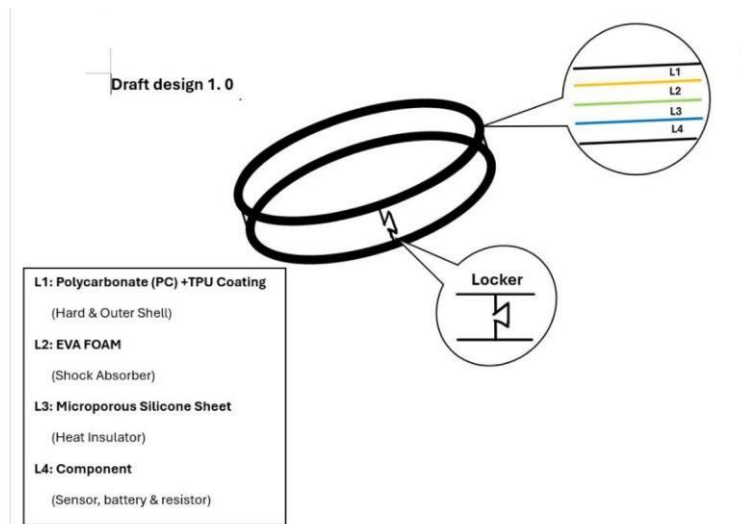
It was an example of how some elderly folk fall through the cracks, unnoticed by others.

The Star reported last week that there were almost 10 cases in the past few months of senior citizens found dead at home where they lived alone.

According to the National Health and Morbidity Survey 2018, an estimated 7.4% of elderly people lived alone.

Context: A case study from Muar illustrating the lethal consequences of the current infrastructure gap. The article highlights that 7.4% of elderly Malaysians live alone and are at high risk of "unnoticed mortality"—a tragedy where the deceased are not found for days. MEMORA's "Tier 2 Escalation Protocol" is specifically designed to eliminate this indignity.

11.4 Appendix D: MEMORA Draft Design 1.0 (Engineering Evolution)



Context: An early schematic demonstrating the foundational "Safety Layering" architecture. This draft highlights the critical inclusion of **Layer 2 (EVA Foam)** for impact absorption during falls and **Layer 3 (Microporous Silicone)** to insulate the skin from battery heat—evidence of our "Dignity-First" engineering priority to prevent user discomfort